
STATISTICAL DISTRIBUTIONS

Theoretical Foundation · Short Notes & Visual Explanations

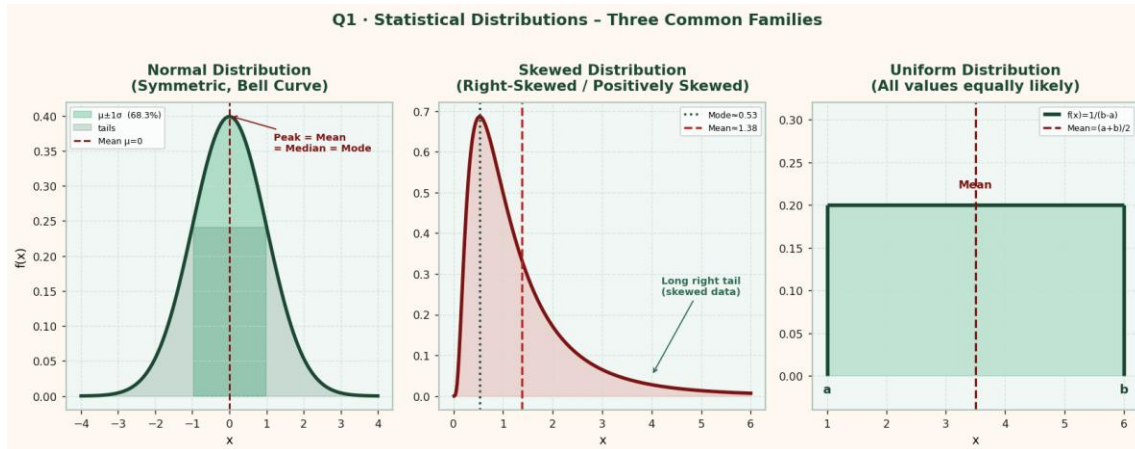
Part A — Data Science & Statistics

Contains 11 Topics · Annotated Diagrams · Formulas · Examples

Q1 What is Statistical Distribution?

A Statistical Distribution describes how values of a variable are spread across a range — it tells us which values are common, which are rare, and the overall pattern of the data.

◆ Three Key Distribution Families



◆ Why Distributions Matter

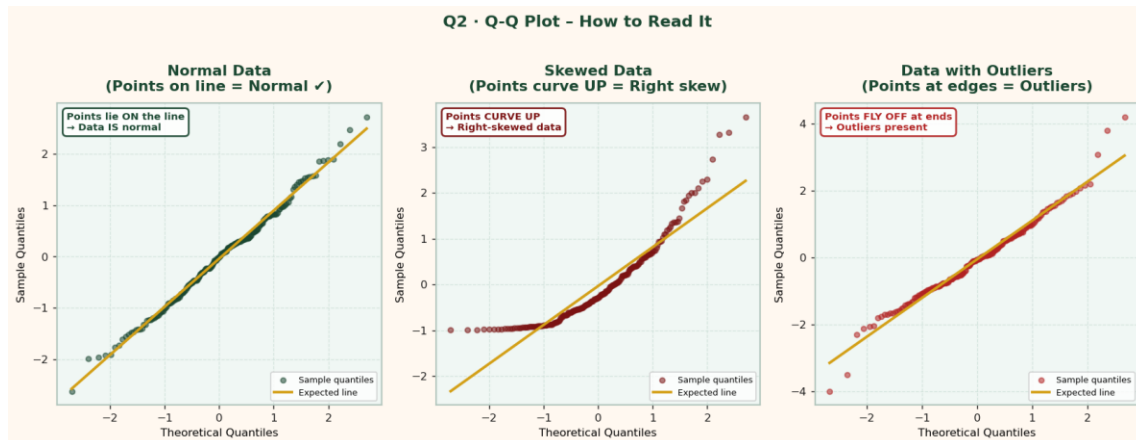
- They describe the pattern of any real-world dataset
- Required to select correct statistical tests (t-test, ANOVA, chi-square)
- Foundation for probability calculations and predictions
- Used in ML for assumptions (Naive Bayes, Linear Regression, etc.)

Distribution Type	Key Characteristic
Normal (Gaussian)	Symmetric bell curve; mean=median=mode
Binomial	Discrete; counts successes in n trials
Poisson	Counts events in fixed time/space
Log-Normal	Skewed; values always positive
Uniform	All outcomes equally likely
Exponential	Time between Poisson events

Q2 What is a Q-Q Plot and Why is it Used?

A Q-Q (Quantile-Quantile) Plot compares the quantiles of your sample data against the theoretical quantiles of a distribution (usually Normal). It is the fastest visual method to check if data follows a specific distribution.

◆ How to Read Q-Q Plots — 3 Cases



◆ Interpretation Guide

Pattern Observed	What It Means
Points lie ON the diagonal line	Data follows Normal distribution ✓
Points CURVE UP at right end	Right-skewed data (positive skew)
Points CURVE DOWN at left end	Left-skewed data (negative skew)
Points FLY OFF at both ends	Outliers or heavy-tailed distribution
S-shaped curve	Lighter tails than Normal

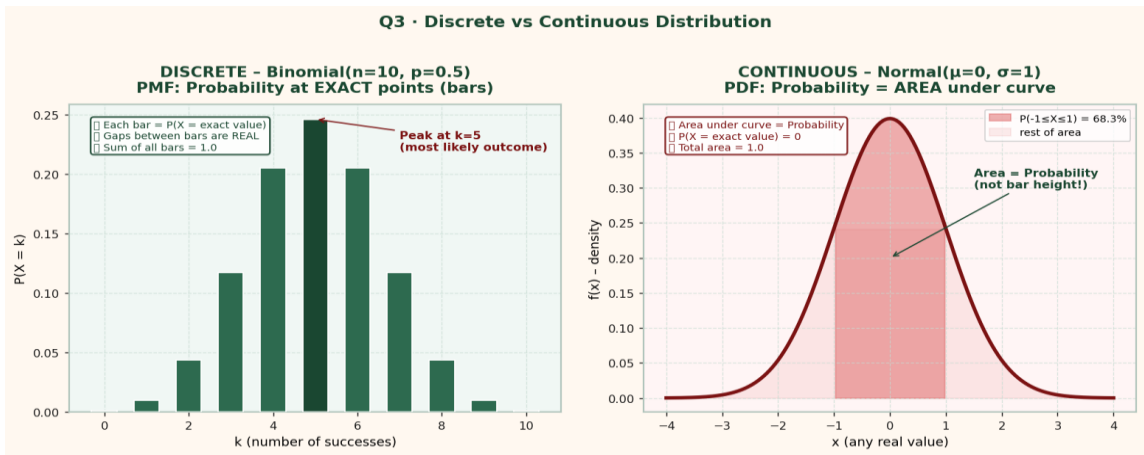
Why Q-Q Plot is Important:

Many tests (t-test, ANOVA, regression) assume normality. If you skip checking this and data is skewed, your test results are **WRONG**. Q-Q plot catches this in seconds — just look at the line!

Q3 Discrete vs Continuous Distributions

The most fundamental classification of distributions is based on what kind of values the random variable can take.

Visual Comparison — Bars vs Smooth Curve



Detailed Comparison

Discrete Distribution	Continuous Distribution
Countable values only: 0, 1, 2, 3...	Any real value in a range: 1.5, 3.14...
Uses PMF (Probability Mass Function)	Uses PDF (Probability Density Function)
$P(X=k)$ has a specific non-zero value	$P(X = \text{exact value}) = 0$ (always!)
Graph: Bar chart with GAPS between bars	Graph: Smooth continuous curve
Sum of all $P(X=k) = 1$	Area under entire curve = 1
Examples: Binomial, Poisson, Bernoulli	Examples: Normal, Log-Normal, Exponential

Memory Trick (Hinglish):

Discrete = Dantey (teeth) — gaps hote hain daanton ke beech. Continuous = Paved road — bilkul smooth, koi gap nahi!

Q4 What is Bernoulli Distribution?

Bernoulli Distribution is the simplest discrete distribution — it models a single experiment (trial) with exactly two possible outcomes: Success ($X=1$) with probability p , and Failure ($X=0$) with probability $(1-p)$.

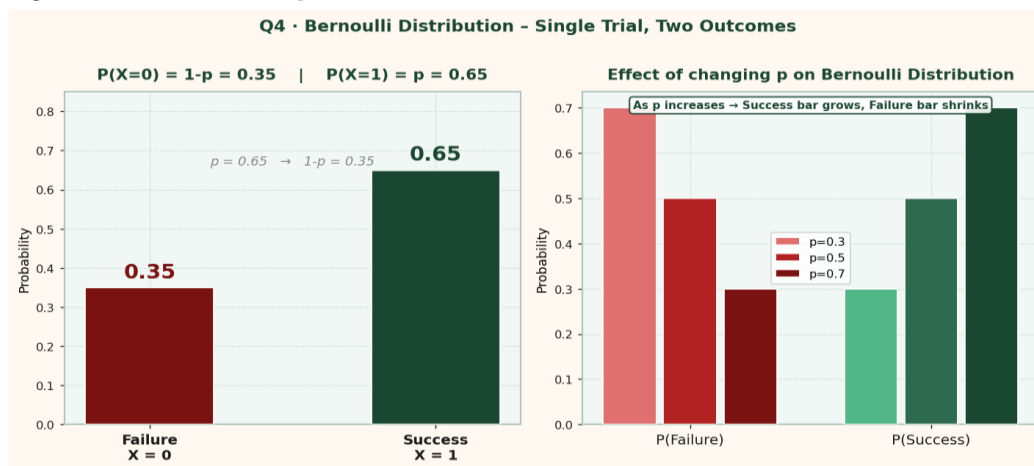
◆ Formula & Parameters

$$P(X = k) = p^k \cdot (1-p)^{1-k} \quad \text{where } k \in \{0, 1\}$$

$$\text{Mean } \mu = p \quad | \quad \text{Variance } \sigma^2 = p(1-p)$$

$$\text{For } p=0.65: \text{ Mean} = 0.65, \text{ Variance} = 0.65 \times 0.35 = 0.2275$$

◆ Probability Bars & Effect of p



◆ Real-World Examples

Scenario	Bernoulli Mapping
Tossing a coin	Heads=1 ($p=0.5$), Tails=0 ($1-p=0.5$)
Spam email detection	Spam=1, Not Spam=0
Medical test result	Positive=1, Negative=0
Customer makes purchase	Buy=1, No Buy=0
Student passes exam	Pass=1 ($p=0.7$), Fail=0 ($1-p=0.3$)

Bernoulli vs Binomial:

Bernoulli = ONE trial only (flip coin once). Binomial = MANY Bernoulli trials combined (flip coin 10 times, count heads). Binomial is just repeated Bernoulli!

Q5 What is Binomial Distribution?

Binomial Distribution models the number of successes (k) in exactly n independent identical Bernoulli trials, each with success probability p . It is one of the most widely used discrete distributions.

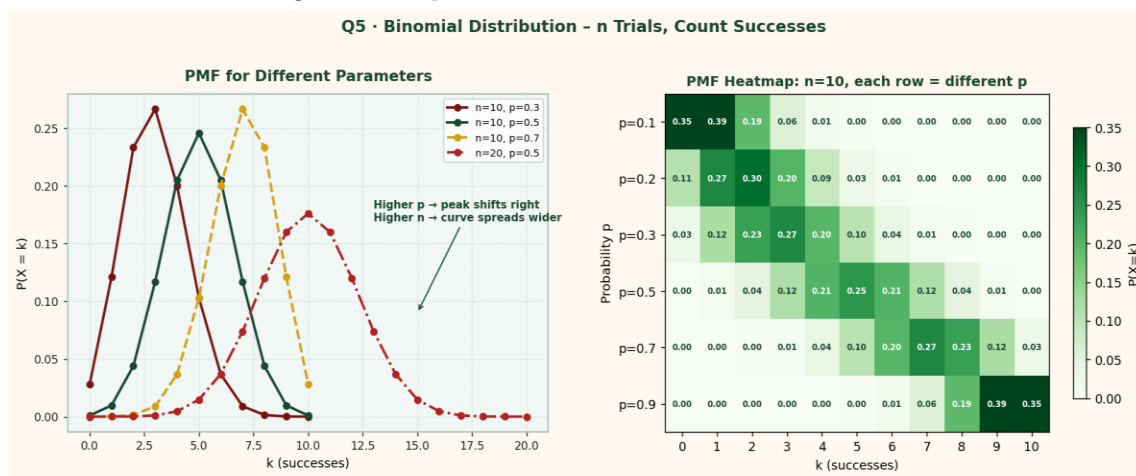
◆ Formula

$$P(X = k) = C(n, k) \cdot p^k \cdot (1-p)^{n-k}$$

$C(n, k) = n! / (k! \times (n-k)!)$ ← number of ways to choose k from n

Mean $\mu = np$ | Variance $\sigma^2 = np(1-p)$

◆ PMF Curves + Probability Heatmap



◆ Conditions for Binomial

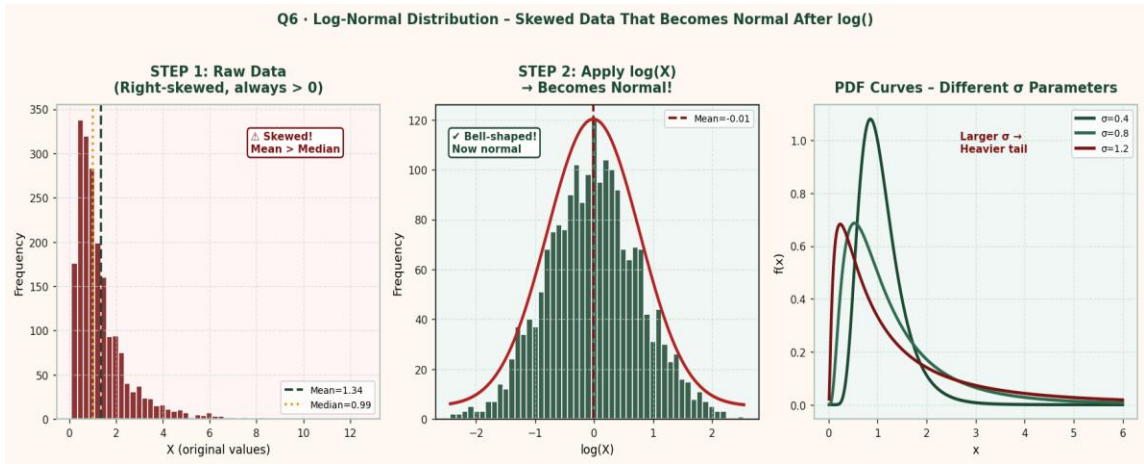
- Fixed number of trials n (decided before experiment)
- Each trial is independent (one result doesn't affect next)
- Only TWO outcomes per trial (Success / Failure)
- Same probability p in every trial

Example Calculation	Answer
$n=10$ coin flips, $P(\text{exactly 6 heads})$?	$C(10, 6) \cdot 0.5^6 \cdot 0.5^4 = 210 \times 0.0156 = 0.205$
$n=5$ exams ($p=0.8$ pass), $P(\text{all pass})$?	$0.8^5 = 0.328 = 32.8\%$
$n=3$ free throws ($p=0.7$), $P(2 \text{ made})$?	$C(3, 2) \cdot 0.7^2 \cdot 0.3 = 3 \times 0.49 \times 0.3 = 0.441$

Q6 Explain Log-Normal Distribution

A variable X follows a Log-Normal distribution if $\ln(X)$ — its natural log — follows a Normal distribution. Log-Normal data is always positive and right-skewed, making it perfect for modeling income, prices, sizes, and growth rates.

◆ Step-by-Step: Skewed → Log Transform → Normal



◆ Formula & Key Properties

$$f(x) = \frac{1}{(x\sigma\sqrt{2\pi})} \cdot \exp\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right] \quad \text{for } x > 0$$

Mean = $\exp(\mu + \sigma^2/2)$ Median = $\exp(\mu)$ Mode = $\exp(\mu - \sigma^2)$

Property	Value / Description
Domain	$X > 0$ only (always positive)
Shape	Right-skewed (positive skew)
After log transform	$\ln(X) \sim \text{Normal}(\mu, \sigma^2)$
Effect of larger σ	Heavier right tail, more skew

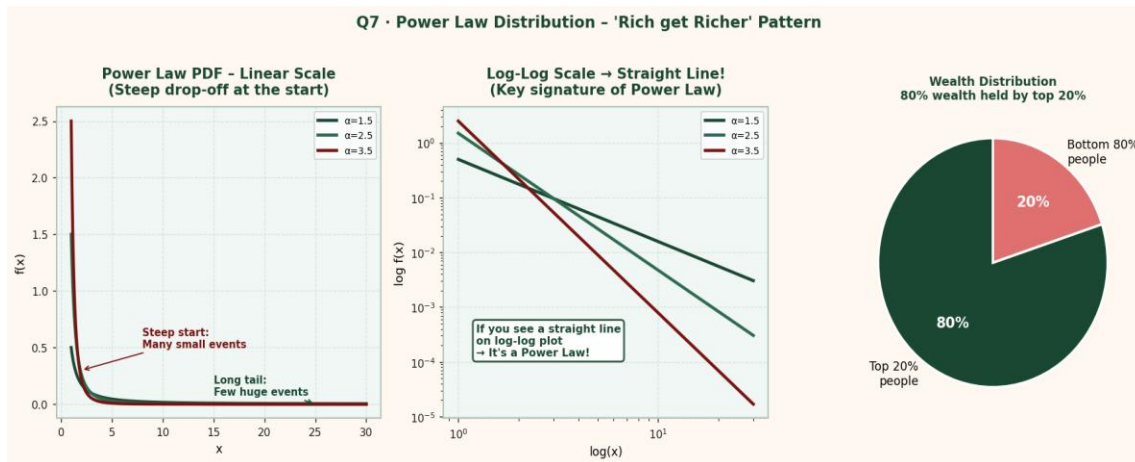
◆ Real-World Applications

- ▶ Stock prices and financial returns (never go below 0)
- ▶ Salary/income distribution across employees
- ▶ Sizes of biological organisms, tumor volumes
- ▶ Website response times and latency
- ▶ Rainfall amounts, earthquake magnitudes

Q7 Explain Power Law Distribution

Power Law Distribution describes phenomena where the probability of an event drops off as a power of its size. It is characterized by a very long heavy tail — meaning rare but extremely large events exist.

◆ Visual — Linear, Log-Log, and Real-World (Pareto 80/20)



◆ Formula

$$P(X = x) \propto x^{-\alpha} \quad (\text{where } \alpha > 1 \text{ and } x \geq x_{\min})$$

KEY TEST: On a log-log plot, Power Law appears as a STRAIGHT LINE with slope = $-\alpha$

α (alpha)	Meaning	Example
$\alpha < 2$	Infinite variance — extreme unpredictability	Internet traffic spikes
$\alpha = 2$	Variance is infinite but finite mean	Word frequencies
$2 < \alpha < 3$	Finite mean, infinite variance	City populations
$\alpha > 3$	Both mean and variance finite	Earthquake magnitudes

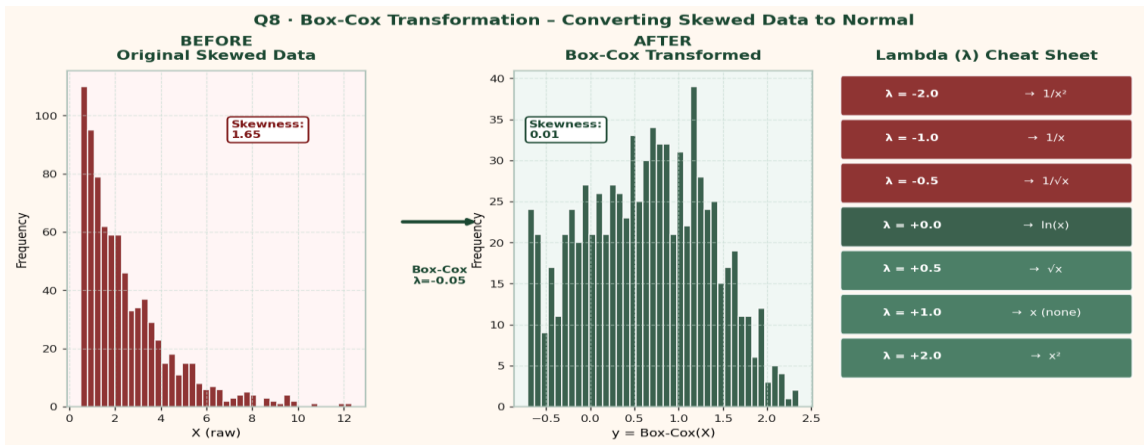
◆ Pareto Principle (80-20 Rule)

- ▶ 80% of wealth held by 20% of people
- ▶ 80% of website traffic goes to 20% of pages
- ▶ 80% of bugs caused by 20% of code
- ▶ Few mega-cities, millions of small towns

Q8 What is Box-Cox Transformation?

Box-Cox Transformation is a power transformation technique that converts non-normal (skewed) data into a Normal distribution by finding the optimal lambda (λ) parameter using Maximum Likelihood Estimation.

◆ Before → Apply λ → After (3-Step Visual)



◆ Formula

$$y(\lambda) = (x^\lambda - 1) / \lambda \quad \text{if } \lambda \neq 0$$

$$y(\lambda) = \ln(x) \quad \text{if } \lambda = 0$$

λ is chosen by Maximum Likelihood Estimation (MLE) — best λ makes data most Normal

Lambda (λ) Value	Transformation Applied
$\lambda = -1$	Reciprocal: $y = 1/x$
$\lambda = -0.5$	Reciprocal square root: $y = 1/\sqrt{x}$
$\lambda = 0$	Natural log (most common): $y = \ln(x)$
$\lambda = 0.5$	Square root: $y = \sqrt{x}$
$\lambda = 1$	No transformation (data already normal)
$\lambda = 2$	Square: $y = x^2$

When to use Box-Cox:

When your data is skewed and you need normality for regression, t-tests, or ANOVA. Box-Cox finds the BEST transformation automatically instead of you guessing log/sqrt.

Limitation: X must be > 0 . For negative values, use Yeo-Johnson transform instead.

Q9 Poisson Distribution with Example

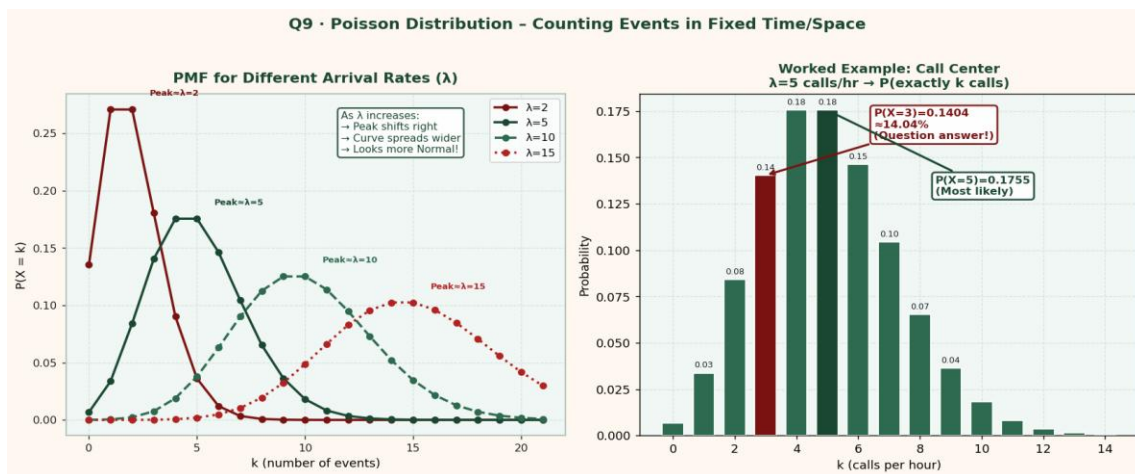
Poisson Distribution models the number of events occurring in a fixed interval of time or space, given that events happen at a known average rate λ and are independent of each other.

◆ Formula

$$P(X = k) = (e^{-\lambda} \cdot \lambda^k) / k! \quad \text{where } e \approx 2.71828$$

Mean $\mu = \lambda$ | Variance $\sigma^2 = \lambda$ (Mean = Variance — unique property!)

◆ PMF Curves + Worked Example Walkthrough



◆ Worked Example — Step by Step

Problem: A call center gets 5 calls/hour on average. Find $P(\text{exactly 3 calls in next hour})$.

Given: $\lambda = 5$ (average rate), $k = 3$ (we want exactly 3)

Formula: $P(X=3) = (e^{-5} \times 5^3) / 3!$

Step 1: $e^{-5} = 0.00674$

Step 2: $5^3 = 125$

Step 3: $3! = 6$

Step 4: $P(X=3) = (0.00674 \times 125) / 6 = 0.8425 / 6 = 0.1404$

Answer: $\approx 14.04\%$ chance of receiving exactly 3 calls.

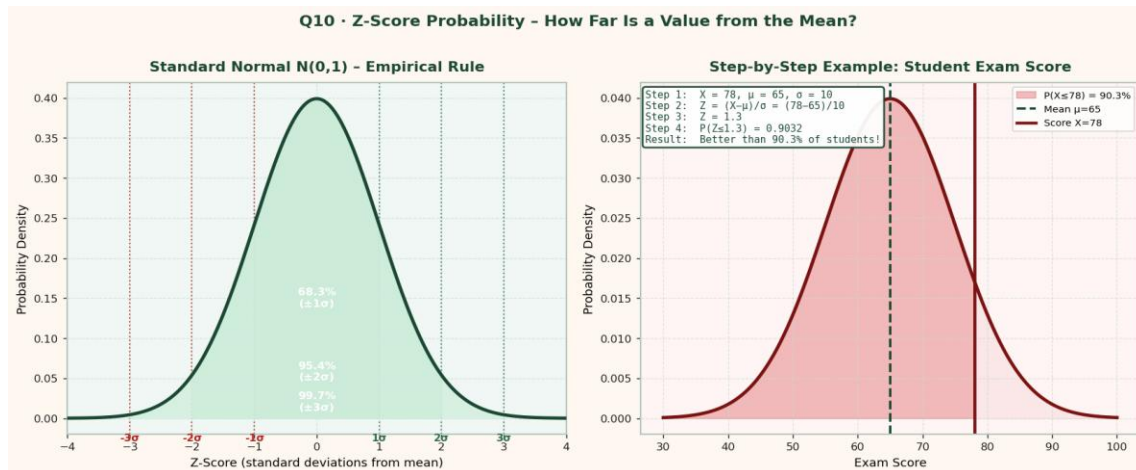
◆ Four Conditions for Poisson

- Events are independent of each other
- Average rate λ is constant over time/space
- Two events cannot happen at the exact same instant
- Events occur one at a time

Q10 What is Z-Score Probability?

A Z-Score standardizes any raw value X by measuring how many standard deviations it sits away from the mean. It converts any Normal distribution into the Standard Normal $N(0,1)$, allowing us to look up probabilities from Z-tables.

◆ Formula + Empirical Rule + Worked Example



◆ Formula

$$Z = (X - \mu) / \sigma$$

X = raw value | μ = population mean | σ = standard deviation

After converting, use Standard Normal Table (Z-table) to find probability

◆ Empirical Rule — 68-95-99.7

Z-Score Range	Probability Covered
-1 to +1 ($\pm 1\sigma$)	68.3% of all data
-2 to +2 ($\pm 2\sigma$)	95.4% of all data
-3 to +3 ($\pm 3\sigma$)	99.7% of all data
$Z > 1.65$	Top 5% (one-tailed test)
$Z > 1.96$	Top 2.5% (95% confidence interval)
$Z > 2.576$	Top 0.5% (99% CI)

◆ Worked Example

Student scores 78 in exam. Class mean = 65, SD = 10.

$$Z = (78 - 65) / 10 = 13/10 = 1.3$$

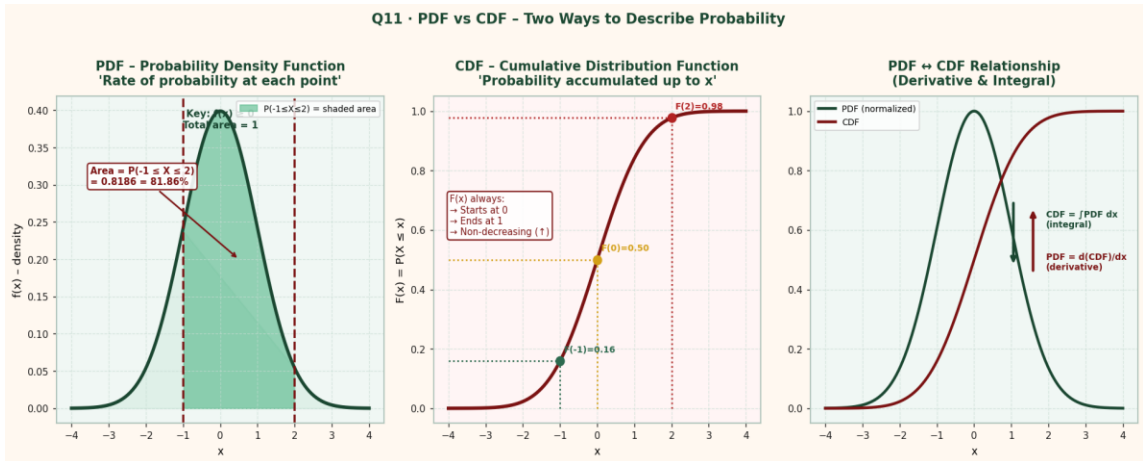
From Z-table: $P(Z \leq 1.3) = 0.9032$

Interpretation: Student scored better than 90.32% of the class!

Q11 PDF vs CDF — Key Differences

PDF (Probability Density Function) and CDF (Cumulative Distribution Function) are the two fundamental mathematical tools for describing continuous probability distributions. They are related by calculus — one is the integral of the other.

Side-by-Side Comparison with Annotations



Detailed Comparison Table

PDF — Probability Density Function	CDF — Cumulative Distribution Function
Notation: $f(x)$	Notation: $F(x) = P(X \leq x)$
Gives the rate/density of probability at x	Gives total probability accumulated up to x
$f(x) \geq 0$ (can exceed 1!)	$0 \leq F(x) \leq 1$ (always between 0 and 1)
$P(X = \text{exact value}) = 0$	$P(X \leq x) = \text{area under PDF up to } x$
Total area under curve = 1	$F(-\infty) = 0$ and $F(+\infty) = 1$
Shape: bell curve, skewed curve, etc.	Shape: S-shaped, always non-decreasing ↑

Mathematical Relationship

CDF from PDF: $F(x) = \int_{-\infty}^x f(t) dt$ (integrate PDF to get CDF)
PDF from CDF: $f(x) = dF(x)/dx$ (differentiate CDF to get PDF)
They carry the same information — just expressed differently!

Practical Use:
PDF → Use when you want to know WHERE probability is concentrated (which values are likely).
CDF → Use when you want to know $P(X \leq \text{some value})$ — like what % scored below 75?

